

US1: Task Progress Tracking

Title: Task Progress Tracking

As a student,

I want to track the progress of my tasks

So that I can keep track of my work progress.

Features

- **Feature #16:** Creates a timeline for the user to assign their tasks into (closed)
- **Feature #17:** Enables user to create personal notes and other kinds of tasks (closed)
- **Feature #23:** Tracking tasks progress (closed)

Technical Requirements

1. Task Progress Tracking

Description: The system must allow students to track the progress of their tasks effectively.

Functional Requirements:

- The user can create personal notes that contain tasks.
- Each task must include a description, due date, and detailed notes.
- Tasks can be organized on a timeline/calendar interface.
- The user can view overall progress for larger tasks, including percentage completion and remaining tasks.

Non-Functional Requirements:

- The application must be user-friendly and intuitive for students.
- The system should be responsive and accessible on both desktop and mobile devices.
- Data must be securely stored and comply with relevant data protection regulations.

Technical Constraints:

- The application will be developed using React for the frontend and Node.js for the backend.
- A MongoDB database will be used to store task and note data.

Dependencies:

- Integration with a calendar API for timeline functionality.
- User authentication must be implemented to ensure secure access to personal notes and tasks.

Potential Risks:

- Inaccurate progress tracking could lead to user frustration and decreased productivity.
- Miscommunication with the outsourced team regarding feature specifications may result in delays.

Acceptance Criteria

- The user can create personal notes that contain their tasks.
- Each task must include a description, due date, and detailed notes.
- Tasks can be organized on a timeline/calendar interface.
- The user can view overall progress for larger tasks, including percentage completion and remaining tasks.

US2: Project Timeline Creation

Title: Project Timeline Creation

As a freelance graphic designer,
I want to create a project timeline
So that I can stay on top of deadlines.

Features

- **Feature #16:** Timeline/calendar (closed)
- **Feature #23:** Tracking tasks progress (closed)

Technical Requirements

1. Project Timeline Creation

Description: The system must allow freelance graphic designers to create and manage project timelines.

Functional Requirements:

- The user can create a new project timeline with a unique title.
- The user can add tasks to the timeline, specifying due dates, priorities (high, medium, low), and notes for each task.
- The user can break down the project into smaller tasks or milestones, with the ability to set dependencies between tasks.

Non-Functional Requirements:

- The system must be responsive and accessible on both desktop and mobile devices.
- The application should load within 2 seconds for optimal user experience.
- Data must be stored securely, adhering to GDPR compliance.

Technical Constraints:

- The application will be built using React for the frontend and Node.js for the backend.
- Database will be managed using MongoDB, ensuring that all task data is stored in a structured format.

Dependencies:

- Integration with a calendar API for due date reminders.
- User authentication must be implemented to ensure that only authorized users can create and manage timelines.

Potential Risks:

- Miscommunication with the outsourced team regarding feature specifications could lead to delays.
- Inadequate testing of the timeline feature may result in bugs that affect user experience.

Acceptance Criteria

- The user can create a new project timeline.
- The user can add tasks to the timeline with due dates, priorities, and notes for each task.
- The user can break down the project into smaller tasks or milestones.

Related Information

- **Missing Acceptance Criteria:** Acceptance criteria for each user story need to be defined, including specific conditions for success and measurable outcomes.
- **Alignment with Standards:** The plan aligns with INVEST by ensuring user stories are independent, negotiable, valuable, estimable, small, and testable. It adheres to ISO-29148:2011 by focusing on clear requirements gathering and stakeholder engagement, ensuring that the user stories are well-defined and actionable.

US3: Advanced Data Analysis for Lecturers

Title: Advanced Data Analysis for Lecturers

As a lecturer,
I want to perform advanced data analysis
So that I can monitor my productivity and student engagement.

Technical Requirements

1. Advanced Data Analysis for Lecturers

Description: The system must enable lecturers to perform advanced data analysis to monitor productivity and student engagement.

Functional Requirements:

- The user can perform data analysis based on events that occurred within a defined time period.
- The system must provide access to relevant data sources, including lecture attendance records, assignment grades, and student participation logs.
- Advanced data analysis algorithms or tools must be implemented to process large datasets efficiently.
- The system should generate comprehensive reports on lecturer productivity metrics, such as time spent on lecture preparation, grading, and student interaction.
- Data analysis results must be visualized using charts, graphs, or dashboards for easy interpretation.

Non-Functional Requirements:

- The application must be user-friendly and intuitive for lecturers.
- The system should be capable of handling large datasets without performance degradation.
- Data must be securely stored and comply with relevant data protection regulations.

Technical Constraints:

- The application will be developed using Python for data analysis and visualization, with a web interface built using React.
- A PostgreSQL database will be used to store data related to lectures, attendance, and student performance.

Dependencies:

- Integration with existing data sources for attendance and grades.
- User authentication must be implemented to ensure secure access to sensitive data.

Potential Risks:

- Inaccurate data analysis could lead to incorrect conclusions about productivity and engagement.
- Integration challenges with existing data sources may delay implementation.

Acceptance Criteria

- The user can perform data analysis based on events that occurred within a defined time period.
- The system must provide access to relevant data sources, including lecture attendance records, assignment grades, and student participation logs.
- Advanced data analysis algorithms or tools must be implemented to process large datasets efficiently.
- The system should generate comprehensive reports on lecturer productivity metrics, such as time spent on lecture preparation, grading, and student interaction.
- Data analysis results must be visualized using charts, graphs, or dashboards for easy interpretation.

US4: Create Personal Notes

Title: Create Personal Notes

As a student,
I want to note down my ideas
So that I can brainstorm using the notes.

Related Information

- **Feature:** #17 (closed) Feature: Create personal notes.

Technical Requirements

1. The system must allow users to create, edit, and delete notes specifically for brainstorming ideas.
2. Implement a search functionality that supports keyword search across all notes with a response time of less than 2 seconds.
3. Ensure that notes are stored in a database with basic security measures, focusing on user accessibility rather than extensive encryption.

Feasibility Feedback

- The user story remains technically feasible with the revised focus on core functionalities. The removal of user authentication simplifies implementation while still allowing for basic security measures.

User Value

- This user story delivers value by enabling students to efficiently capture and organize their ideas, facilitating better brainstorming and study practices.

Acceptance Criteria

- The user is able to create and manage notes.
 - Enable quick search functionality to find specific notes or ideas.
 - Ensure data security and privacy measures to protect sensitive information.
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US5: Item Categorization

Title: Item Categorization

As a student,

I want to create different categories

So that I can categorize my notes.

Related Information

- **Feature:** #19 Feature: Item categorization.

Technical Requirements

1. The system must allow users to create, edit, and delete categories for organizing notes.
2. Users should be able to assign multiple categories to a single note.
3. Implement a linking function that allows users to connect related categories.
4. Develop a global search/filter/sort functionality that enables users to find notes based on various criteria such as name, date, and topic, with a response time of less than 2 seconds.
5. Ensure that the categorization system is intuitive and user-friendly, minimizing the risk of category assignment errors.

Feasibility Feedback

- The user story is technically feasible, with clear requirements that can be implemented within the existing framework. However, attention must be given to the user interface design to ensure ease of use and to mitigate the risk of category assignment errors.

User Value

- This user story delivers value by enabling students to efficiently organize their notes, facilitating better study practices and information retrieval.

Acceptance Criteria

- The user can define different categories (e.g., study, work).
 - The user can assign their notes to the category they want.
 - Each note can belong to multiple different categories.
 - The product has a function to link categories.
 - The product has a global search/filter/sort function to find notes by name/date/topic/etc.
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US6: Add Tasks to Journal

Title: Add Tasks to Journal

As a student,

I want to add my own task into the journal

So that I can track my weekly schedule.

Technical Requirements

1. The system must allow users to add individual tasks as todo/schedule items with a title, description, and due date.
2. Implement a timeline/calendar view that supports daily, weekly, and monthly displays of tasks.
3. Users should be able to drag and drop tasks onto the timeline for easy scheduling.
4. The system must allow tasks to toggle between different states: 'todo', 'doing', and 'done'.
5. Implement a date and time picker for scheduling tasks.
6. Develop a reminder functionality that sends notifications for upcoming tasks.

Feasibility Feedback

- The user story is technically feasible within the current technology stack (Next.js, AWS Lambda, DynamoDB). The requirements are clear and can be implemented effectively. However, attention must be given to the user interface design to ensure it is intuitive and user-friendly, especially for the timeline/calendar view.

User Value

- This user story delivers value by enabling students to effectively manage their tasks and schedules, enhancing their productivity and organization.

Acceptance Criteria

- The user can add tasks with a title, description, and due date.
 - The user can view tasks in a timeline/calendar format.
 - The user can drag and drop tasks onto the timeline.
 - The user can toggle tasks between 'todo', 'doing', and 'done'.
 - The user can use a date and time picker for scheduling tasks.
 - The user receives reminders for upcoming tasks.
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US7: Lecture and Assignment Schedule Management

Title: Lecture and Assignment Schedule Management

As a lecturer,

I want to plan and manage my lectures/assignments schedule for multiple courses

So that I can ensure a smooth academic semester.

Related Information

- **Features:**

- #5 Feature: Scheduling
- #16 Feature: Timeline/calendar (closed)
- #23 Feature: Tracking tasks progress (closed)

Product Details

- **Input Options:**
 - Course name
 - Course code
 - Course description

Technical Requirements

1. The system must support the creation and management of schedules for multiple courses, allowing lecturers to input course details (name, code, description) and lecture specifics (dates, times, topics, locations).
2. The scheduling interface must be user-friendly and responsive, ensuring accessibility across devices.
3. Notifications for upcoming lectures and assignment deadlines must be configurable by the user, with options for email and in-app alerts.
4. The backend must support data persistence for all scheduling information, utilizing a relational database to store course and schedule data.
5. The system must implement user authentication to verify the identity of lecturers and authorization to grant access to their schedules, ensuring that only authorized users can modify their schedules.
6. The application should be able to handle concurrent users without performance degradation, ensuring that multiple lecturers can manage their schedules simultaneously.

Acceptance Criteria

1. Ability to create and manage schedules for multiple courses within the same platform.
 2. Input options for course details such as course name, code, and description.
 3. Capability to add lecture dates, times, topics, and locations for each course.
 4. Notifications or reminders for upcoming lectures and assignment deadlines.
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US8: Task and Notes Timeline View

Title: Task and Notes Timeline View

As a student,

I want to view my tasks and notes in a timeline

So that I know what to do each day.

Technical Requirements

1. The system must provide a timeline view that displays tasks and notes in chronological order, allowing students to see their daily responsibilities at a glance.
2. The timeline view must be interactive, enabling students to click on tasks or notes for more details or to edit them.
3. The interface must be user-friendly and responsive, ensuring accessibility across various devices (desktop, tablet, mobile).
4. The backend must support data retrieval for tasks and notes, utilizing a relational database to store user data securely.
5. The system must allow for filtering and sorting of tasks and notes based on criteria such as due date, priority, or category.
6. The application should leverage existing note functionalities to display notes within the timeline view, ensuring that any updates to notes are automatically reflected in the timeline without requiring additional integration.

Potential Risks

- Performance issues with the timeline view when handling a large number of tasks and notes.
- Ensuring data integrity during retrieval and display, particularly with real-time updates.

User Value

This user story delivers value by providing students with a clear and interactive way to manage their tasks and notes, helping them stay organized and focused on their daily responsibilities. The timeline view enhances usability and engagement, making it easier for students to track their progress and meet deadlines.

Acceptance Criteria

1. The timeline view displays all tasks and notes in chronological order.
 2. Students can interact with tasks and notes to view details or edit them.
 3. The interface is responsive and accessible on various devices.
 4. Users can filter and sort tasks and notes based on due date, priority, or category.
 5. The system leverages existing note functionalities to display notes within the timeline view, ensuring that any updates to notes are automatically reflected.
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US9: Search for Notes with Customizable Filters

Title: Search for Notes with Customizable Filters

As a student,

I want to easily search for my notes with customizable filters

So that I can find what I need to.

Related Information

- **Stakeholders:** Students, Educators, Administrators
- **Objectives:** Improve note accessibility and organization for students.

Product Details

- **Product Name:** Note Management System
- **Version:** 1.0

Technical Requirements

1. The system must implement a search algorithm that can filter notes based on date, subject, and keywords.
2. The database must support indexing on the fields used for filtering to ensure search results are returned within a maximum response time of 1 second under normal load conditions.
3. The user interface must allow students to select and save filter preferences, which should be stored in the user's profile.
4. Ensure compliance with data privacy regulations when storing user preferences.

Feasibility Feedback

- The user story remains technically feasible with the updated requirements. The change to a 1-second response time is achievable with proper indexing and optimization of the search algorithm. The removal of the irrelevant requirement regarding testing under various load conditions simplifies the implementation process.

Acceptance Criteria

- The system must allow students to filter notes by date, subject, and keywords.
- The search functionality should return results within a maximum response time of 1 second under normal load conditions.
- Users should be able to save their filter preferences for future searches.

User Value

- This user story delivers significant value by allowing students to efficiently search for their notes using customizable filters, which enhances their study experience and productivity. The technical requirements ensure that the feature is user-friendly and performs optimally, meeting the needs of the stakeholders.
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US10: Customize Profile with Rewards

Title: Customize Profile with Rewards

As a student,

I want to customize my profile with rewards from completing tasks

So that I can stand out from other users.

Technical Requirements

1. The system must allow users to select and display rewards earned from completed tasks on their profile.
2. Implement a user-friendly interface for profile customization that allows users to easily navigate and select rewards.
3. Provide a variety of rewards for users to choose from, including badges and items from a point shop.
4. Ensure that any changes made to the profile are saved in real-time and reflected immediately on the user's profile.
5. Integrate the profile customization feature with the existing gamification elements, ensuring that rewards are linked to task completion and user engagement.

Feasibility Feedback

- The user story is technically feasible within the current technology stack (Next.js, AWS Lambda, DynamoDB). The requirements are clear and can be implemented effectively. However, attention must be given to the user interface design to ensure it is intuitive and user-friendly.

User Value

- This user story delivers value by allowing students to personalize their profiles, enhancing their engagement and motivation through visible rewards for their achievements.

Acceptance Criteria

- The user can select and display rewards earned from completed tasks on their profile.
- The profile customization interface is user-friendly and intuitive.
- Users can choose from a variety of rewards, including badges and items from a point shop.
- Changes to the profile are saved in real-time and reflected immediately.